



Kalman filter for mobile-robot attitude estimation: Novel optimized and adaptive solutions

Ákos Odry^{a,*}, Róbert Fullér^{b,c}, Imre J. Rudas^{b,d}, Péter Odry^a

^a Department of Control Engineering and Information Technology, University of Dunaújváros, Dunaújváros, Hungary

^b Institute of Applied Mathematics, Óbuda University, Budapest, Hungary

^c Department of Informatics, Széchenyi István University, Győr, Hungary

^d University Research, Innovation and Service Center, Óbuda University, Budapest, Hungary



ARTICLE INFO

Article history:

Received 20 October 2017

Received in revised form 22 March 2018

Accepted 24 March 2018

Available online 30 March 2018

Keywords:

Adaptive filter

Attitude determination

Filter tuning

Inertial measurement unit

Kalman filter

Sensor fusion

ABSTRACT

This paper proposes two novel approaches to estimate accurately mobile robot attitudes based on the fusion of low-cost accelerometers and gyroscopes. The first part of the paper demonstrates the use of a special test bench that both enables simulations of various dynamic behaviors of wheeled robots and measures their real attitude angles along with the raw sensor data. These measurements are applied in a simulation environment and we outline an offline optimization of Kalman filter parameters. The second part of the paper introduces a novel adaptive Kalman filter structure that modifies the noise covariance values according to the system dynamics. The instantaneous dynamics are characterized regarding the magnitudes of both the instantaneous vibration and the external acceleration. The proposed adaptive solution measures these magnitudes and utilizes fuzzy-logic to modify the filter parameters in real time. The results show that the adaptive filter improves the overall filter convergence by a remarkable 10.9% over using the optimized Kalman filter, thereby demonstrating its efficacy as an accurate and robust attitude filter. The proposed filter performances are also benchmarked against other common methods indicating that the flexibility of the developed adaptive filter allowed it to compete and even outperform the benchmark filters.

© 2018 Elsevier Ltd. All rights reserved.

1. Introduction

1.1. Attitude estimation

Micro-electro-mechanical systems (MEMS), such as gyroscopes and accelerometers, are characterized by low costs, low power consumptions, and small sizes, which make them popular candidates for implementation in embedded systems. These sensors are generally applied to solve the relative localization problem, where the positions and orientation information of moving objects are determined using a MEMS-based inertial measurement unit (IMU) composed of the aforementioned MEMS sensors. The MEMS-based localization approach has been widely investigated and is widely utilized in robotics [1–7], human position tracking [8,9], medical fields [10], sports equipment [11,12], and automated driving [13–17] both in industry and in scientific research. Independent of the specific application, it is always advantageous to

* Corresponding author at: University of Dunaújváros, Tánács Mihály u. 1, 2400 Dunaújváros, Hungary.

E-mail addresses: odrya@uniduna.hu (Á. Odry), fuller.robert@nik.uni-obuda.hu (R. Fullér), rudas@uni-obuda.hu (I.J. Rudas), podry@uniduna.hu (P. Odry).

combine the accelerometer and gyroscope sensors, utilizing their individual features to provide accurate and robust orientation results.

Gyroscopes measure angular rates. Therefore, the orientation of a moving object can be obtained via numerical integration. However, the output of an MEMS gyroscope contains both temperature dependent bias and noise, which inevitably cause cumulative errors (boundless drift). This error renders attitude calculation unviable, especially for long-term measurements. Accelerometers provide measurements that are complementary to the angular rate measurements. These sensors can be used as inclinometers, providing accurate orientation results even over long periods. However, accelerometers only produce reliable orientation-related sensor data when they are in stationary states (there is no external acceleration or vibration). This means that the decomposition of the pure gravity vector can be used to determine the orientation. Conversely, in a dynamic situation, the external acceleration cannot be distinguished from the gravity vector. Therefore, the effects of external acceleration drastically decrease the reliability of accelerometer-based orientation calculation.

Due to the aforementioned difficulties, various sensor fusion techniques that combine gyro-based high frequency attitude estimation with the rough, low-frequency attitude correction (drift cancellation) provided by the accelerometer have been developed. Among these sensor fusion techniques, the Kalman filter (KF) and its variant for nonlinear cases, the extended KF (EKF), are among the solutions most often applied to MEMS-IMU based attitude estimation. These algorithms are recursive, optimal state estimators under the minimum mean square error (MMSE) criterion for Gaussian stochastic dynamic systems. In this paper the authors investigate state estimation, based on signals corrupted by Gaussian noises. Therefore, the paragraphs below offer a detailed discussion of the techniques applied for performance enhancement of (E) KF-based state estimation, but for the sake of comprehensiveness, the penultimate paragraph briefly mentions the so-called maximum correntropy filtering, developed recently for state estimation of non-Gaussian stochastic systems.

The KF produces an optimal filter performance, if the characteristics of the modeled noise in the Gaussian stochastic system are known completely (and modeled by candidate noise covariance values). However, in most cases the noise statistics of the analyzed state space model cannot be determined. This is an even more critical problem for MEMS-IMU based orientation calculations of moving objects, since neither the external accelerations nor vibrations are deterministic, resulting in radical measurement noise that cannot be modeled appropriately. Therefore, the covariance matrices are usually selected on an ad hoc basis via time-consuming trial-and-error procedures performed based on engineering intuition. This iterative process represents a compromise between the accuracy and filter dynamics in which the ultimately determined noise covariance values both roughly describe the measurement noise and cover the model approximations.

To overcome the difficulties related to trial-and-error tuning and to achieve superior filter performances, there are two common approaches in the literature. On the one hand, the filter convergence can be enhanced by numerical optimization (usually performed offline). To optimize, an environment is created (with the assistance of other sensors or filters) in which the true state can be measured along with the IMU data. By evaluating the performance index, the KF noise covariance values are tuned with an optimization algorithm. This procedure, along with the application of the downhill simplex algorithm, is explained in detail through various theoretical examples in [18]. Similarly, a neural-network based approach to tuning the noise statistics was investigated in [19]. Kownacki [20] dealt with the filtration of accelerometer and gyroscope signals. Separate KFs were designed and tuned based on the assumption that the sensors were in stationary states (no external acceleration was present). The noise covariance values were tuned with the simplex search method through the minimization of a defined cost function. The differential evolution [21] and genetic algorithms (GAs) [22] have also been applied to the optimization of extended KF (EKF) covariance matrices.

Other filter performance enhancement techniques tune the KF matrix covariance values adaptively, usually based on the magnitude of external (dynamic) acceleration. The dynamic acceleration is defined as the difference between the magnitudes of the accelerometer reading and the gravity vector. In both [23,24], an EKF with the aforementioned adaptation technique was proposed for a miniature attitude and heading reference system (AHRS) in which different measurement noise variances are taken into account based on the identified stationary-, low-, and high-acceleration modes (i.e., threshold-based switching of covariance matrices). A similar adaptation approach to estimating lower trunk orientations was investigated by Mazza et al. [8]. Moreover, both the threshold levels and the noise statistics were optimized using the PatternSearch algorithm. Suh [7] describes an adaptive KF solution for preventing vibrations from affecting the attitude estimations of unmanned aerial vehicles (UAVs), whereas Jurman et al. [25] proposed adaptation laws for both the process and measurement noise covariance values. Lee et al. [26] investigated an acceleration-model based approach to attitude estimation in which the dynamic acceleration is also estimated and utilized to compensate for the external acceleration in the attitude determination process. Furthermore, this approach was compared with the threshold-based switching technique, but it did not demonstrate significant improvements in estimation accuracy over the threshold-based technique. Gośliński et al. [27] proposed an adaptive KF for estimating human body orientation during walking based on the data acquired from inertial sensors of a present-day smartphone. In the adaptation law, the noise covariance matrix was modified based on the variance of input signal, moreover, the upper and lower bounds of covariance values were determined with the aid of optimization. The proposed filtering technique was benchmarked against the Android OS estimation algorithm and a complementary filter consisting of first order low-pass and high-pass filters [28].

It is worth mentioning that the estimation performance of KF is likely to deteriorate considerably, if signals are corrupted by non-Gaussian impulsive noises [29]. The maximum correntropy KF (MCKF) and its constrained state estimator variant (MCKF-SC) efficiently estimate the states of a non-Gaussian stochastic system by using the maximum correntropy criterion (MCC) instead of the MMSE in the development of filter equations [30,31]. Additionally, an improved MCKF and its

square-root implementations were established in [32]. However, the comprehensive examination on a nonlinear state-estimation case study of radar tracking problem in non-Gaussian noise environment has not confirmed the superior performance of the MCC-based techniques [33]. Correntropy criterion-based KF for orientation estimation was also investigated in [34]. These recent developments in the realm of Kalman-type filters are among the important techniques to be utilized in state estimation of non-Gaussian stochastic dynamic systems.

Comprehensive overviews of (nonlinear) attitude estimation solutions are provided in [35,36].

1.2. Two-wheeled inverted pendulum systems

This paper focuses primarily on one-dimensional attitude (tilt angle) estimation of two-wheeled inverted pendulum (TWIP) systems. However, both the problems examined and solutions proposed can be applied universally to every mobile system (e.g., wheeled/legged robots or unmanned aerial vehicles).

A TWIP system consists of two actuated wheels and an inner body that forms a pendulum. The control objective is to simultaneously guarantee the planar motion of the wheels and stabilize the pendulum around its unstable upright position. Therefore, providing accurate attitude values as input to the applied control structure is essential for stabilizing the unstable system.

Because the relative orientation of a TWIP body cannot be measured with encoders, its attitude is usually estimated by a trial-and-error tuned KF [4–6]. Trial-and-error KF tuning is a challenging issue for TWIPs because they are naturally unstable. Moreover, the implemented KF and controller are linked in a closed loop. Therefore, it is difficult to determine whether a badly tuned controller or attitude estimator results in unsatisfactory system behavior or not. Additionally, there are two main types of disturbances that cause the TWIP system attitude estimation to become unreliable: external acceleration and external vibrations. External acceleration can occur as a result of either a predefined planar motion (i.e., following a desired trajectory) or external influences (i.e., the robot is pushed or collides with obstacles). Vibrations also arise during closed-loop behavior, because real systems encounter driving mechanism backlash that produces unwanted system behavior, especially when larger control signals are applied [37]. Therefore, closed-loop performance usually depends on both ad hoc KF tuning (with a virtually unknown convergence quality) and a controller that roughly tolerates estimation that is inaccurate, noisy, or delayed.

1.3. Contributions of the paper

The discussion in the previous section illustrates that the procedure for selecting optimal filter parameters for TWIPs, thus providing an optimal filter convergence, remains an important issue. Moreover, the effects of inherent external acceleration and vibrations are not usually taken into account explicitly, and, to our knowledge, no study has yet investigated whether considering the magnitudes of these disturbances can improve filter robustness and accuracy.

This paper addresses the preceding issues and proposes two distinct approaches to attitude estimation. The first approach describes a procedure for overcoming the compromise solution related to ad hoc KF tuning by finding the optimal KF filter parameters, while the second approach introduces a novel adaptive KF structure that modifies the filter parameters based on external disturbances, thus achieving a superior filter performance. The main contributions of this paper are as follows.

1. Introducing a test bench that enables simulations of various (accelerating and non-accelerating) system behaviors as well as measurement and qualification of the filter convergence.
2. Applying particle swarm optimization to KF noise covariance values and achieving an optimized filter performance.
3. Introducing two methods for measuring the magnitudes of external disturbances (i.e., instantaneous vibrations and acceleration).
4. Developing a novel adaptive KF structure that fuses the magnitudes of the disturbances together and utilizes fuzzy-logic based heuristic IF-THEN rules that modify the noise covariance values based on the dynamic behavior.
5. Comparing the proposed filter performances to two popular AHRS algorithms (i.e., a gradient-descent based algorithm and a nonlinear complementary filter) proved that the proposed solutions are competitive and even outperform these common methods.

The test bench can also be utilized to tune and optimize other attitude filters (as it is demonstrated in Section 6). Moreover, the proposed optimized and adaptive solutions are universally applicable to any robotic application in which attitude estimation is required and dynamic effects (such as vibration and acceleration) influence the filter performance significantly.

The remainder of the paper is organized as follows. Section 2 introduces mathematical models related to MEMS IMU measurements and formulates a two-dimensional KF. Section 3 describes both the TWIP system and an applied test environment in which measurements can be collected under various dynamic conditions. In Section 4, we discuss a procedure for optimizing the KF noise covariance values. Section 5 describes methods for measuring external disturbances and elaborates on the adaptive KF approach. In Section 6, the filter performances are compared against other common orientation estimation methods. Finally, Section 7 provides our conclusions and recommendations for future studies.

2. Attitude estimation with low cost MEMS-IMUs

As mentioned in the introduction, neither the accelerometer nor the gyroscope is capable of providing accurate, stable attitude values. Therefore, the purposes of attitude estimation are to integrate and process raw measurements and to provide attitude estimates that are smoothed properly. A KF is a recursive, state-space model-based algorithm that both enables the integration of the aforementioned sensors and provides the optimal state estimation based on the properly defined noise covariance matrices (related to sensor noise).

2.1. Gyroscope model

The raw gyroscope measurement Ω_k^B consists of three main components: the true angular velocity ω_k^B , the non-static bias term $\omega_{0,k}$, and the additive measurement noise μ_k . It is given by

$$\Omega_k^B = \omega_k^B + \omega_{0,k} + \mu_k, \quad (1)$$

where the superscript B denotes vectors expressed in the body frame and k is a discrete-time variable. In Eq. (1), the additive rate noise is assumed a zero-mean, white, Gaussian variable with the following characteristics:

$$E[\mu_k] = 0, \quad E[\mu_k \mu_l^T] = \sigma_\mu^2 \delta_{kl}, \quad [\sigma_\mu^2] = \left(\frac{\text{deg}}{\text{s}}\right)^2 \frac{1}{\text{Hz}}, \quad (2)$$

where σ_μ^2 denotes the noise variance and δ_{kl} is the Kronecker delta. Moreover, the term $\omega_{0,k}$ in Eq. (1) is considered a slowly varying bias (as a result of temperature sensitivity) modeled by the random walk process

$$\omega_{0,k} = \omega_{0,k-1} + \eta_k, \quad (3)$$

where η_k denotes driving Gaussian noise with a σ_η^2 noise variance, i.e.,

$$E[\eta_k] = 0, \quad E[\eta_k \eta_l^T] = \sigma_\eta^2 \delta_{kl}, \quad [\sigma_\eta^2] = \left(\frac{\text{deg}}{\text{s}}\right)^2 \text{Hz}. \quad (4)$$

After being reformulated from the gyroscope measurement (Eq. (1)), the true angular velocity, can be integrated numerically to compute the orientation using $\theta_{k+1} = \theta_k + \omega_k T_s$, where T_s is the sampling time. However, due to the bias term $\omega_{0,k}$ and the presence of the measurement noise μ_k , the gyro-based realization of θ_k produces an unbounded accumulation error. Therefore, gyroscope measurements yield only short-term accuracy (i.e., high-frequency attitude realization can be achieved with rate gyro measurements).

2.2. Accelerometer model

The raw accelerometer output A_k^B consists of four main components: the external acceleration α_k^B , the contribution of gravitational acceleration g_k^B , the bias term a_0 , and the additive measurement noise v_k , and it is given by

$$A_k^B = (\alpha_k^B - g_k^B) + a_0 + v_k. \quad (5)$$

The bias term is usually compensated for during the calibration procedure [38], although the process for achieving this exceeds the scope of this article. If a robot (or another mobile mechatronic system) is in stationary state (i.e., no dynamic acceleration is occurring and $\alpha_k^B \approx 0$), then the gravity vector can be used to calculate the raw attitude realization $\theta_{A,k}$ [24] with

$$\theta_{A,k} = \arctan2(A_{y,k}, A_{z,k}) = \theta_k + v_k, \quad (6)$$

where $A_{y,k}$ and $A_{z,k}$ are the instantaneous measurements of the accelerometer. This realization, which is described by Eq. (6), can also be considered the sum of the real attitude θ_k and additive noise v_k that represents the effects of v_k from Eq. (5) after the trigonometric function has been evaluated. In a manner similar to the gyroscope model, v_k is assumed to be Gaussian white noise with the following characteristics:

$$E[v_k] = 0, \quad E[v_k v_l^T] = \sigma_v^2 \delta_{kl}, \quad [\sigma_v^2] = \text{deg}^2 \frac{1}{\text{Hz}}. \quad (7)$$

Based on Eqs. (5) and (6), we can deduce that if there is an external acceleration ($\alpha_k^B \neq 0$), then the attitude realization given by Eq. (6) provides unreliable results and a drastically reduced accuracy because the ratio of $A_{y,k}$ to $A_{z,k}$ does not provide relevant attitude information (i.e., the pure gravity vector cannot be applied). In our implementation, the external acceleration is not modeled explicitly. Instead, its effects are absorbed by v_k (i.e., a significantly larger σ_v^2 noise variance is expected when $\alpha_k^B \neq 0$). Therefore, the accelerometer measurements yield adequate attitude realizations (so-called low-frequency attitude realizations) only when the system is in a non-accelerating mode.

2.3. Sensor fusion with Kalman filter

The previously described approaches can be synthesized to utilize the advantages of both types of sensors and to obtain attitude results with higher reliabilities and accuracies. In such a synthesis, the gyro-based attitude realization is extended by the attitude realization derived from the accelerometer data. Therefore, the unbounded integration error is compensated for and long-term stability is achieved. To fuse these sensors, a linear KF is implemented.

This KF is a recursive algorithm that provides an optimal estimation $\hat{x}$ of the noisy state vector x (whose dynamics are described by a state-space equation) such that:

$$\begin{aligned} E[x_k - \hat{x}_k] &= 0, \\ E[(x_k - \hat{x}_k)(x_k - \hat{x}_k)^T] &\rightarrow \inf. \end{aligned} \quad (8)$$

When MEMS IMU data are considered, the state dynamics are constructed from both the discrete-time integrated true angular velocity (ω_k reformulated from Eq. (1)) and the random walk process of the bias term (Eq. (3)). This means that the state propagation is described by the state vector $x_k = (\theta_k, \omega_{0,k})^T$, the input variable $u_k = \Omega_k$, and the process noise vector $w_k = (\mu_k, \eta_k)^T$ as

$$\begin{aligned} x_{k+1} &= \Phi x_k + \Gamma u_k + w_k, \\ \begin{bmatrix} \theta \\ \omega_0 \end{bmatrix}_{k+1} &= \begin{bmatrix} 1 & -T_s \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \theta \\ \omega_0 \end{bmatrix}_k + \begin{bmatrix} T_s \\ 0 \end{bmatrix} \Omega_k + \begin{bmatrix} \mu \\ \eta \end{bmatrix}_k, \end{aligned} \quad (9)$$

where Φ is the state transition matrix, Γ denotes the input matrix, and $T_s = 1/f_s$ is the sampling time. Based on Eq. (6), the measurement equation is formed (as an updated absolute orientation) with an output of $z_k = \theta_{A,k}$, measurement noise of v_k , and an output matrix of H as

$$\begin{aligned} z_k &= Hx_k + v_k, \\ \theta_{A,k} &= [1 \quad 0] \begin{bmatrix} \theta \\ \omega_0 \end{bmatrix}_k + v_k. \end{aligned} \quad (10)$$

According to the stochastic hypothesis, if the process and measurement noise vectors w_k and v_k are uncorrelated ($E[w_k v_k^T] = 0$) and modeled with zero-mean, white, Gaussian random variables (as was assumed in both the gyroscope and accelerometer models represented by Eqs. (2), (4), and (7)), then the Kalman filter provides an optimal estimation with a minimum state-vector variance. The recursive algorithm uses the state-space Eqs. (9) and (10) along with the covariance matrices of w_k and v_k given by $Q = E[w_k w_k^T]$ and $R = E[v_k v_k^T]$, respectively, to propagate the states, process the measurements, and update the covariance estimates in the time and measurement update equations [39]. Namely, the time update equations determine the *a priori* state estimate ($\hat{x}_k^-$) and estimate error covariance (P_k^-) as:

$$\begin{aligned} \hat{x}_k^- &= \Phi \hat{x}_{k-1} + \Gamma u_{k-1}, \\ P_k^- &= \Phi P_{k-1} \Phi^T + Q, \end{aligned} \quad (11)$$

where $\hat{x}_{k-1}$ and P_{k-1} denote the *a posteriori* (updated) state estimate and estimate error covariance at step $k - 1$. The measurement update equations are responsible for correcting the *a priori* estimates employing both the measurement z_k and its noise covariance R . First, the Kalman gain is calculated:

$$G_k = P_k^- H^T (H P_k^- H^T + R)^{-1}, \quad (12)$$

then, both the state estimate and the estimate error covariance are updated (i.e., the *a posteriori* state vector and error covariance matrix are determined):

$$\begin{aligned} \hat{x}_k &= \hat{x}_k^- + G_k (z_k - H \hat{x}_k^-), \\ P_k &= (I - G_k H) P_k^-, \end{aligned} \quad (13)$$

where I denotes the 2×2 identity matrix.

Because the matrices Φ , Γ , and H in Eqs. (9) and (10) contain constants, the estimation (or filter) performance is directly influenced, and thus determined, by the respective choices of the process and measurement noise covariances Q and R as well as the initial value of the estimation error covariance $P_0 = E[(x(0) - x_0)(x(0) - x_0)^T]$. The following sections propose a solution based on numerical optimization for automatic tuning of these parameters. Moreover, we describe an adaptive KF approach that varies the aforementioned covariance values based on the dynamical behavior of the system.

3. Test environment

To evaluate the state estimation error, measure the filter performance, and realize the optimization of the KF parameters, the true state (i.e., the true attitude of the TWIP body θ_k) must be known. Our proposed test bench provides a set of special circumstances to the TWIP that allow the real attitude angle to be measured and the accelerometer and gyroscope measurements to be collected.

3.1. Electro-mechanical structure of the test bench

Due to the mechanical structures of TWIP systems, the pendulum (the body of the robot) is typically realized between the stators and the rotors of their applied motors. The relationships between the angular positions of the wheels ($\theta_{w,k}$) and the pendulum (θ_k) are given by

$$\theta_{r,k} = \theta_{w,k} - \theta_k, \quad (14)$$

where $\theta_{r,k}$ is the relative motor angle measured by the incremental encoders attached to the motor shafts. If the wheels are prevented from rotating (i.e., $\theta_{w,k} = 0$), then, based on Eq. (14), the rotary encoders can be used to measure the true angular position of the TWIP body θ_k .

The test bench depicted in Fig. 1 takes advantage of this condition by using two (shaft-clamping) jaws to pin down the wheel shafts and prevent their rotation. The test bench jaws are attached to a movable plate that slides back and forth on two parallel 400-mm long rails via linear bearings. The position of the plate is measured by the attached encoder. This electro-mechanical structure enables external acceleration to occur simultaneously with the inner body's oscillation, allowing a variety of dynamic (vibrating and accelerating) system behavior to be simulated and measured. Fig. 1 shows both a photograph and the CAD model of the test environment. The shafts of the robot are fixed between the jaws. In the CAD model, the side- and top-printed TWIP circuit boards are set to invisible to reveal the inner workings of the robot.

The embedded TWIP system is described in [40,41]. Its DC motors (model No. Faulhaber 1024N003S) are equipped with both planetary gearheads of 64 reduction ratio and two-channel incremental encoders of 100 lines per revolution resolution (model No. PA2-100). Thus, the accuracy of true attitude (ground truth) measurements employing X4 encoding is:

$$\Delta_\theta = \frac{360^\circ}{4 \times 100 \times 64} \approx 0.0141^\circ. \quad (15)$$

During the measurement, the DC motors are driven using different sinusoidal signals (varying both amplitude and frequency), resulting in a wide variety of TWIP body oscillations. Simultaneously, the movable plate is slid back and forth, simulating various horizontal acceleration values. The encoder measurements attached to the motor (the true attitude) and the sliding plate (the true horizontal acceleration), along with the instantaneous accelerometer and gyroscope data, are collected and sent to the PC for further evaluation. A three-channel incremental encoder (model No. HEDS-5540 A11) of 500 lines per revolution resolution is fixed to a wheel of $r = 19$ -mm radius (see Fig. 1). Therefore, the accuracy of horizontal position measurement employing X4 encoding is:

$$\Delta_y = \frac{2\pi r}{4 \times 500} \approx 0.0597 \text{ mm}. \quad (16)$$

3.2. Measurement results

Dynamic behaviors, including stationary states, various angular velocities, mild and intense vibrations, and mild and intense acceleration were measured and recorded. The measurement lasted for about 350 s, and the simulated dynamic behavior was characterized using the following ranges: 0 – 6.5 Hz for body oscillation frequency, ± 800 deg/s for angular velocity, and ± 1.5 g for external horizontal acceleration. Fig. 2 depicts two parts of the whole measurement in which the angle realization quality can be observed under both static and diverse dynamic conditions. The entire measurement data set can be accessed in the supplementary online material [42].

The first and second rows of Fig. 2 compare the real attitude (measured by the motor encoder) with the attitude realizations provided by the accelerometer and gyroscope data, while the third row shows the external acceleration (measured by the plate encoder) applied during the measurement process. On the one hand, due to the integration of the bias and noise components, the gyro-based angle realization was characterized by unbounded drift (e.g., the accumulated error was about 15° after 100 s and approximately 40° after 300 s). On the other hand, for low-frequency oscillations with no external acceleration, the accelerometer-based realizations describe the instantaneous oscillations accurately (e.g., at around 80 s). However, when either external acceleration or high angular velocity is present, the quality of the accelerometer-based realization decreases drastically (e.g., at around 75 s and 85 s). In fact, the instantaneous realizations are essentially useless in these cases (as shown on the entire right side of Fig. 2).

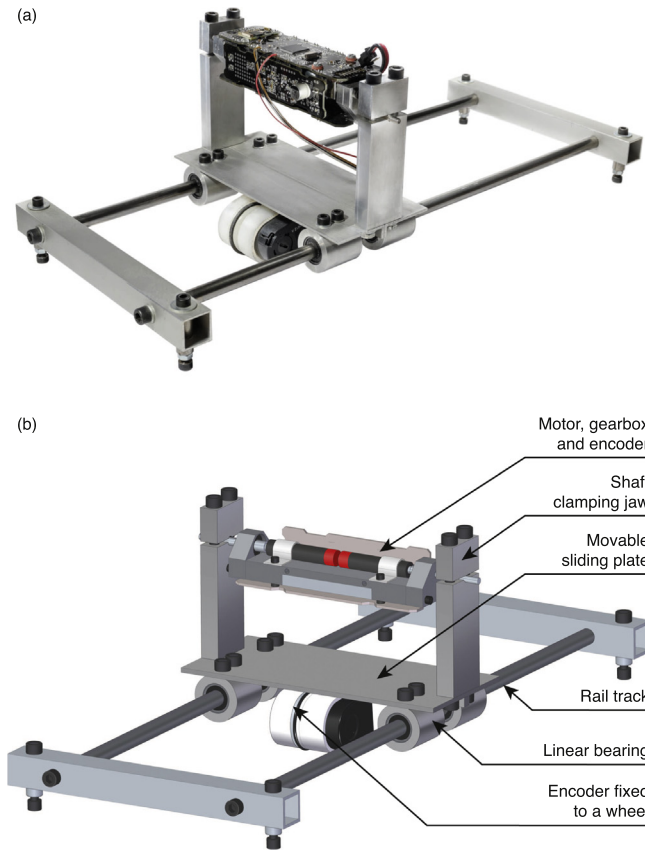


Fig. 1. Photograph and CAD model of test bench.

4. Optimization of filter parameters

Because the embedded TWIP system is characterized by a low processing speed and the recorded measurements were made available in MATLAB, the KF coefficient optimization was performed offline in a MATLAB/Simulink simulation environment. The block diagram of the measurement process and the applied optimization procedure is depicted in Fig. 3. The inputs of the simulation environment are the real angular position (i.e., the encoder measurement of θ_k) and the realized angular position and velocity values (i.e., $\theta_{A,k}$ and Ω_k calculated from the IMU measurements), while its output was the estimation error $e_k = \theta_k - \hat{\theta}_k$, where $\hat{\theta}_k$ denotes the output of the implemented KF algorithm. The KF parameter tuning was performed by defining a fitness function quantifying the estimation quality and applying a method for optimizing the noise covariance values by minimizing the fitness function value.

4.1. Initialization of filter parameters

The initial KF parameter values need to be defined in the simulation environment. Such initialization is necessary for the respective process and measurement noise covariance matrices Q and R , the state vector of the state Eqs. (9) and (10) $\hat{x}_0$, and the estimation error covariance matrix P_0 .

The process noise values μ_k and η_k in Eq. (9) were considered to be statistically independent (i.e., uncorrelated), as it is usually assumed [20,26,27,35]. Therefore, a diagonal process noise covariance matrix (Q), along with the measurement noise variance value (R), was investigated in the optimization procedure, specified as:

$$Q = \begin{bmatrix} q_{00} & 0 \\ 0 & q_{11} \end{bmatrix}, \quad R = \rho. \quad (17)$$

q_{00} , q_{11} , and ρ were introduced to provide simpler notation for the optimization. The noise variance initialization is typically determined via Allan variance analysis for a stationary system state [38,43,44]. However, if the system is operating in

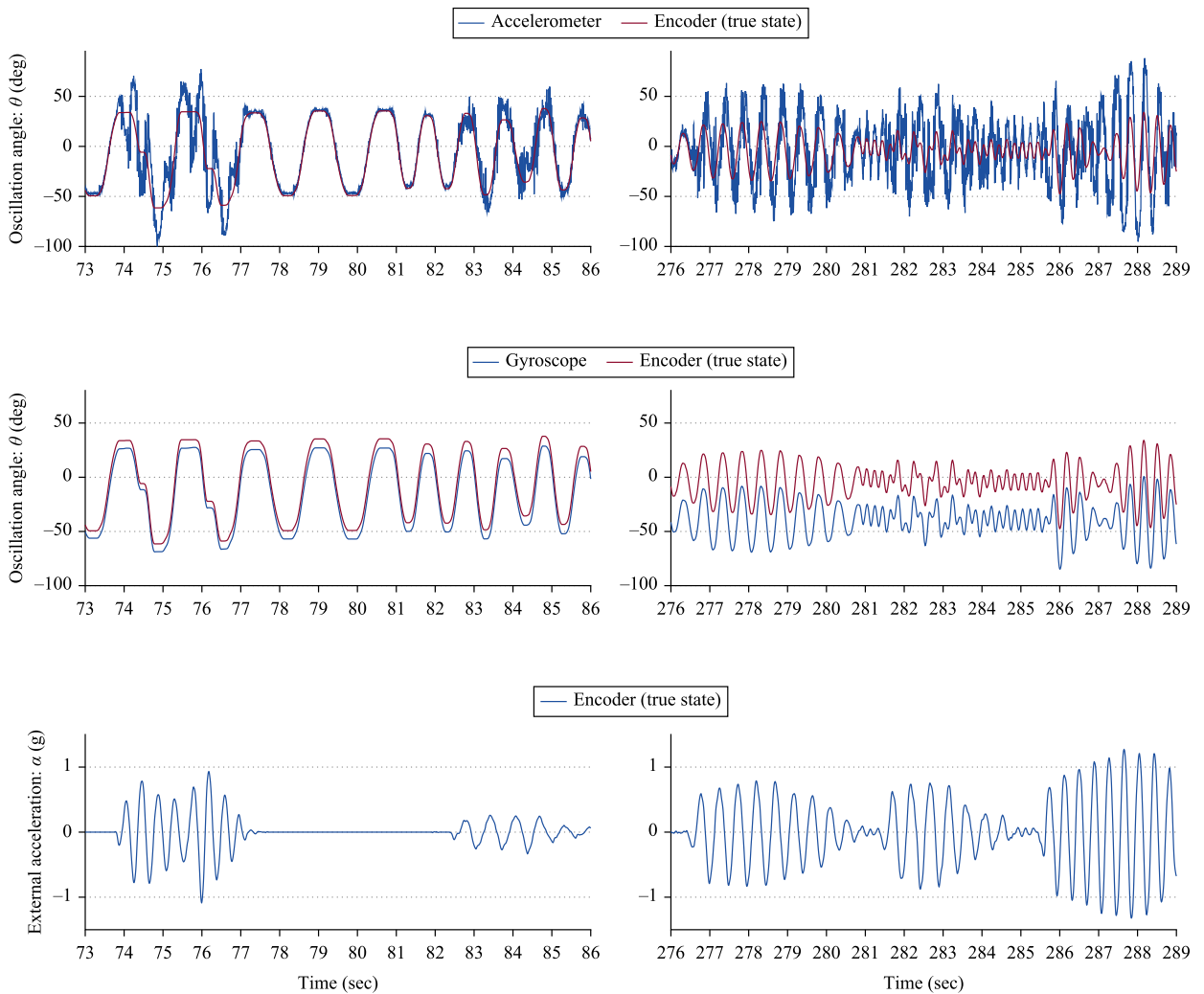


Fig. 2. Two time slots from the 350 s long measurement.

dynamic mode, then the Allan variance initialization method loses validity, especially with respect to ρ , which absorbs and represents the effects of external acceleration (Eq. (6)) as well as the sensor noise.

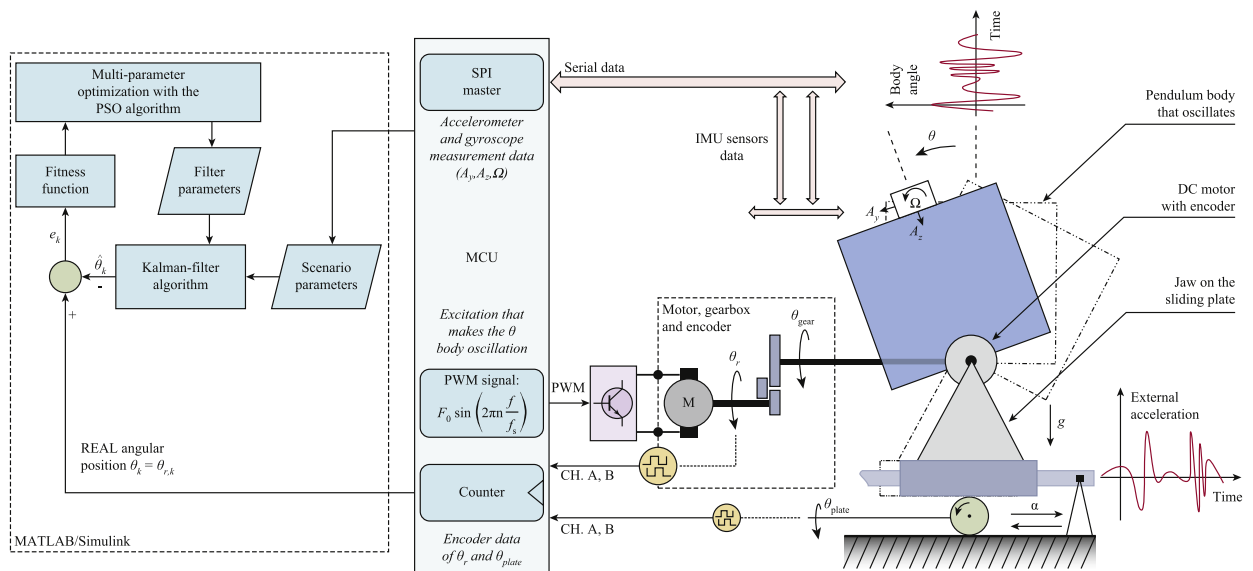
One advantage of this optimization process is that the noise analyses can be omitted, since rational initial guesses for q_{00} , q_{11} , and ρ will cause these parameters to converge to the optimal possible noise variances in the parameter space. Therefore, the following values were chosen for the initialization of noise variances: $q_{00} = 5 \cdot 10^{-7}$, $q_{11} = 5 \cdot 10^{-9}$, and $\rho = 10^5 \cdot q_{00} = 0.05$, where the multiplication factor 10^5 indicates that the realizations of the accelerometer-based attitudes ($\theta_{A,k}$) are expected to be significantly more unreliable than gyro-based angular velocity realizations (Ω_k) due to the system's dynamic behavior.

When the measurements began, the TWIP body angle was 0° . Therefore, we chose the initial state vector $\hat{x}_0 = (0, 0)^T$ and the initial value of state covariance matrix $P_0 = \mathbf{O}_{2 \times 2}$. The estimation error covariance is updated continuously in KF algorithms, thus the initial P_0 guess affects the overall KF performance only slightly. Considering this, the elements of P_0 were not optimized.

4.2. Fitness function

The KF performance was measured by a fitness function that quantified the differences between the real and estimated state values.

In most practice-oriented complex problems, the mean absolute error (MAE) or the mean squared error (MSE) are chosen as the fitness function in the model parameter optimization procedure. On one hand, because these formulas can be easily implemented and evaluated (even in embedded systems) and are well-suited to measurement uncertainties, many research



Based on the foregoing facts, the mean square error (MSE) was selected to be the fitness function:

$$F = \frac{1}{N} \sum_{k=1}^N e_k^2 = \frac{1}{N} \sum_{k=1}^N (\theta_k - \hat{\theta}_k)^2, \quad (18)$$

Since we assumed that the measurement errors in Eqs. (9) and (10) are independent, normally distributed random variables, the optimization of the noise variances by minimizing F in Eq. (18) will correspond to the so-called maximum likelihood estimate [56].

4.3. Particle swarm optimization

Among the heuristic optimization algorithms, particle swarm optimization (PSO) was selected as the most suitable for this problem, because it has demonstrated greater effectiveness than genetic algorithms and other heuristic methods [1,51,57,58]. In addition, PSO is a population-based search algorithm that uses the fitness function to guide the search in the search space; therefore, unlike gradient-based optimization methods, the PSO does not have difficulties with nonlinear, noisy, or discontinuous functions and is less susceptible to becoming trapped in local minima. These advantages are discussed in detail in [46].

PSO utilizes individual particles that form a swarm, imitating the swarm behavior of flocking birds, to search for the global minimum or maximum within a search space. During the particles' flights, they adjust their positions based on both their own experiences and the experiences gained by the swarm as a whole. Specifically, each particle broadcasts its current optimum local points to its neighboring particles. Therefore, each particle knows not only its own optimal position but also the optimal positions of its neighbors as well as the optimal position achieved by the swarm as a whole (the current global optimum). These identified optimal positions are then used by the swarm as reference points for the search process in the iteration's next step [59]. Let X_i and V_i denote the position and velocity vector of the i th particle in the swarm, while P_i and G indicate the personal best position (which gives the best fitness value so far) of the i th particle and the global best position

achieved (i.e., the position of the best individual), respectively. The velocity and position vectors are modified in every generation based on the following equations:

$$\begin{aligned} V_i^{k+1} &= \varpi V_i^k + c_1 r_1 (P_i^k - X_i^k) + c_2 r_2 (G^k - X_i^k), \\ X_i^{k+1} &= X_i^k + V_i^{k+1}, \end{aligned} \quad (19)$$

where c_1 and c_2 are positive constants, $r_1, r_2 \in [0, 1]$ are random values, and ϖ is the inertia weight. These parameters were selected as $\varpi = 0.9$, $c_1 = 0.5$ and $c_2 = 1.5$ based on previous studies [1,47,57]. In the present study, we used the Particle Swarm toolbox for MATLAB [60] to implement the algorithm.

4.4. Results

Once the noise variance ranges were defined and the corresponding fitness function was formulated (Eq. (18)), the optimization algorithm could begin running. We chose $n_{gen} = 30$ and $n_{pop} = 30$ for the numbers of generations and populations, respectively, because the optimization problem itself is characterized by only three parameters (i.e., ρ , q_{00} , and q_{11}). The ranges of the parameters were defined heuristically. The algorithm was run twice in succession, and its ranges and initial values were redefined based on the subresults of the first iteration. The third column of Table 1 summarizes the results of the parameter optimization process, which are discussed in the rest of this subsection.

The optimization process used the initial values defined in the previous subsection. An intense external acceleration was applied during the measurement of the dynamic behavior, and its effect were absorbed in the measurement noise v_k (Eq. (6)). Therefore, the noise variance ρ was expected to converge to a higher value. Indeed, the optimization converged to a noticeably higher variance value (up from 0.05 to 0.35) that more thoroughly represented the noise characteristics of the accelerometer-based realizations. At the same time, the process noise variances converged to notably smaller values (from $5 \cdot 10^{-7}$ to $6.38 \cdot 10^{-8}$ for q_{00} and from $5 \cdot 10^{-9}$ to $3.08 \cdot 10^{-12}$ for q_{11}), meaning that the optimization converged such that the noisy state propagation Eq. (9) became much more reliable but the measurement update equation more uncertain. Owing to the lower process noise covariance Q , the filter gain decreased. Therefore, the optimization resulted in a KF characterized by dynamics slower than those of the algorithm using the initial values. Because the dynamic behavior we simulated and measured on the test bench covered both static and extreme (vibrating and accelerating) TWIP conditions, the faster KF dynamics were not expected to be necessary. Considering the above, the optimization provided satisfactory results. If the filter dynamics are too slow for certain applications, then the subresults of an earlier optimization run could be applied.

Fig. 4 depicts the results of the parameter optimization process. In the first rows, the estimation performances are shown both before (by the red curves) and after (by the yellow curves) the optimization, and the estimated values are compared to the true attitudes. In the second rows, the estimation errors of the initial (blue curves) and optimized (red curves) KF are highlighted. A noticeable performance improvement can be observed in the plots. The curves for the optimized parameters fit both frequencies and amplitudes of the true body attitudes to a satisfactory degree. The fitness function value is significantly improved by the parameter optimization. Specifically, it decreased from the initial $F_{init} = 8.6921$ (filter performance achieved with the initial filter parameters) to $F_{opt} = 1.9077$, indicating that a 78% filter performance improvement was achieved with the proposed optimization procedure.

The effects of the external acceleration and vibration still decreases the estimation quality drastically (e.g., around 75 s on the left side and the whole right side of Fig. 4) resulting in an unsatisfactory initial KF performance (red curves, before optimization). However, a combination of optimized filter parameters was found that enabled the KF to effectively suppress the effects of these external disturbances, and to provide both satisfactory estimation accuracy and fast filter convergence for the intended application.

5. Adaptive Kalman filter approach

The previous section demonstrated that, with the assistance of optimization, a combination of the noise covariance values Q and R can be found such that the resulting state estimation performance is satisfactory. However, the optimized Q and R

Table 1
Initial and optimized values with optimization bounds.

Symbol	Initial	Optimized	min	max
<i>First run: $F_{init} = 8.6921 \rightarrow F_{opt} = 2.3447$</i>				
ρ	0.05	0.27905	0.008	0.3
q_{00}	$5 \cdot 10^{-7}$	$1.08 \cdot 10^{-7}$	$1 \cdot 10^{-7}$	$2 \cdot 10^{-6}$
q_{11}	$5 \cdot 10^{-9}$	$1.17 \cdot 10^{-10}$	$1 \cdot 10^{-10}$	$2 \cdot 10^{-8}$
<i>Second run: $F_{init} = 2.3252 \rightarrow F_{opt} = 1.9077$</i>				
ρ	0.27	0.35459	0.25	0.45
q_{00}	$1 \cdot 10^{-7}$	$6.38 \cdot 10^{-8}$	$7 \cdot 10^{-9}$	$7 \cdot 10^{-7}$
q_{11}	$1 \cdot 10^{-10}$	$3.08 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$4 \cdot 10^{-10}$

The bold values indicate the optimized filter parameters.

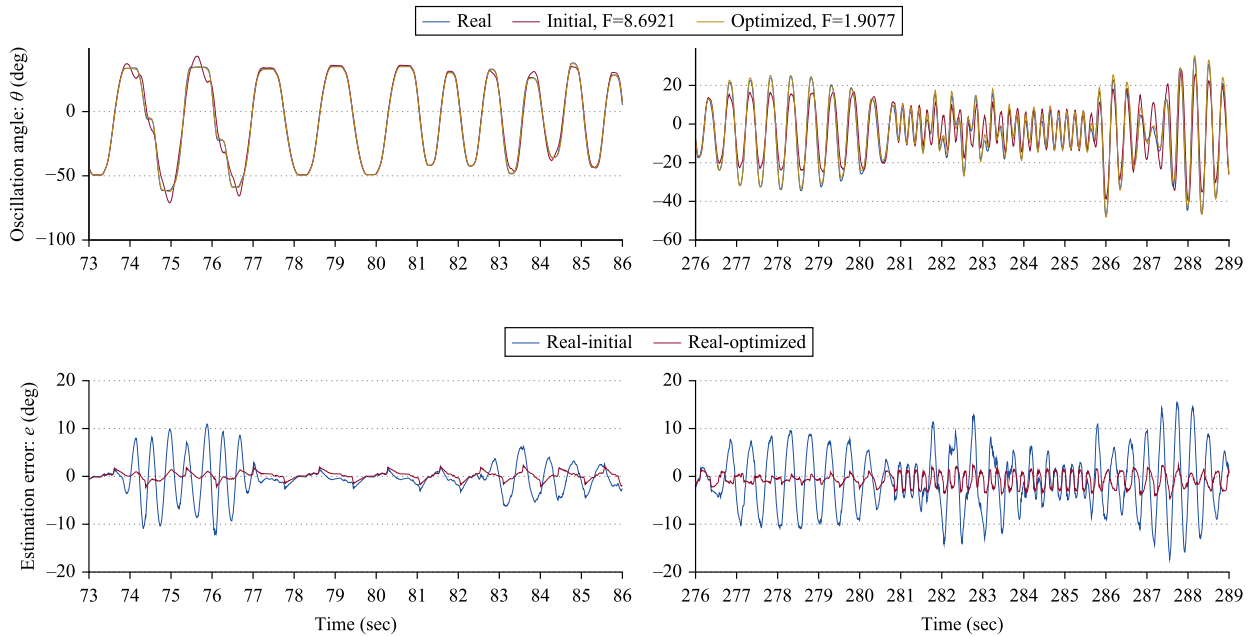


Fig. 4. Initial and optimized Kalman filter performances compared to true body attitudes.

matrices do not characterize accurately the real noise existing in the system but rather represent broad variance values that cover the whole noise and the model approximations in the measurement whether the system is stationary or subject to intense external disturbances. In the present study, this outcome was expected because the state-space equation did not model the external acceleration. Instead, the effects of the external disturbances were absorbed in the assumed white noise component of Eq. (10).

Because the KF performance is primarily influenced by the noise covariance values Q and R , we investigated whether an adaptive approach that varied these matrices according to the instantaneous dynamical behavior could provide a performance superior to that of the filter convergence in the previous section or not. The instantaneous dynamical behavior is characterized by two factors: the magnitudes of the vibration and the external acceleration. If these factors are described by relevant measures in real time, then online manipulation of the noise covariance values can be realized and the estimation performance can be enhanced.

In the following subsections, we describe an adaptive KF approach that both measures the aforementioned external disturbances and modifies the noise variance assumed in the measurement model via a fuzzy inference machine. The results demonstrate that the proposed approach further improves the state estimation quality.

5.1. Measuring vibration magnitude

The magnitudes of instantaneous system vibrations can be described by the oscillation frequency of the TWIP body. Among the two MEMS sensors, the gyroscope provides reliable measurements of the angular velocity for both low and high oscillations. Therefore, the gyroscope data can be utilized to estimate the instantaneous oscillation frequency.

Using this approach, the TWIP body oscillation frequency was estimated with a fast Fourier transform (FFT)-based evaluation of short gyro measurement packets. The high sampling frequency of the sensor allowed us to gather measurement packets of length L . Through the FFT algorithm's evaluation of these short measurements, an estimation of the oscillation frequency ($\hat{f}$) was made. The estimation algorithm functioned as follows.

1. Collect a data packet x of length L from the gyroscope measurements. The value of L depends on the application requirements, since it is a trade-off between the amount of information used in the FFT calculation and the estimation delay. Larger L values will provide finer oscillation spectra but longer estimation delays ($d = L/f_s$). In our study, the sampling frequency was 800 Hz and the window size was $L = 400$ ($d = 0.5$ s delay).
2. Compute a discrete Fourier transform of the data packet x to obtain frequency domain information about the instantaneous oscillation. The output of the FFT algorithm is represented by the ordered pair $(f_i, |\Omega_i|)$, with f_i and $|\Omega_i|$ representing the frequency components and their corresponding amplitudes, respectively. Namely,

$$\mathcal{W}_l = \sum_{k=0}^{L_{\text{FFT}}-1} x_k e^{-j\left(\frac{2\pi l k}{L_{\text{FFT}}}\right)}, \quad l = 0, \dots, L_{\text{FFT}} - 1 \quad (20)$$

and

$$(f_i, |\Omega|_i) = \left(\frac{f_s i}{L_{\text{FFT}}}, \frac{2}{L} |\mathcal{W}_i| \right), \quad i = 0, \dots, \frac{L_{\text{FFT}}}{2}, \quad (21)$$

where L_{FFT} denotes the length of the transform. We selected $L_{\text{FFT}} = 2^9$, yielding an estimation resolution of $\frac{f_s}{L_{\text{FFT}}} = 1.5625$ Hz.

3. Estimate the oscillation frequency (expected to be below $f_{\text{thr}} = 8$ Hz in our case) by finding the highest-intensity frequency component that is smaller than f_{thr} with

$$\begin{aligned} f_{\text{max}} : (f_{\text{max}}, |\Omega|_{\text{max}}) \wedge |\Omega|_{\text{max}} = \max_{\forall i, f_i \leq f_{\text{thr}}} |\Omega|_i, \\ \hat{f} = \begin{cases} 0, & \text{if } |\Omega|_{\text{max}} < |\Omega|_{\text{thr}} \\ f_{\text{max}}, & \text{otherwise} \end{cases} \end{aligned} \quad (22)$$

where, along with the conditions $f \leq f_{\text{thr}}$ and $|\Omega|_{\text{max}} < |\Omega|_{\text{thr}}$, the frequency components related to the measurement noise are isolated (the high frequency components with small magnitudes). In the implementation of the algorithm, we chose $|\Omega|_{\text{thr}} = 10$ deg/s.

4. Repeat the steps starting from 1.

Fig. 5 demonstrates the results of using the aforementioned algorithm on four different data packets. In the first rows, the collected oscillations are shown in the time domain, while the second rows depict the corresponding FFT algorithm results. Evaluation of the algorithm's third step (Eq. (22)) results in the oscillation frequency estimate ($\hat{f}$) depicted near the peak of the curves. The embedded necessary condition in Eq. (22) is explained for the fourth case (on the right side) of Fig. 5 in which no oscillation occurs and the FFT result contains only the corresponding noise spectrum. When f_{thr} is not introduced, the 106-Hz frequency component would correspond to the maximum intensity and result in an incorrect estimate, while the introduction of $|\Omega|_{\text{thr}}$ prevents the incorrect selection of the 3-Hz component as the estimated frequency as well. Fig. 6 highlights the real-time algorithm evaluation over a section of the 350 s measurement. The blue curve represents gyro-based angular rate measurements and the red curve shows the estimation of the oscillation frequency based on data packets of length $L = 400$. The figure illustrates that the aforementioned FFT-based algorithm provides information related to the instantaneous vibrations of a dynamical system.

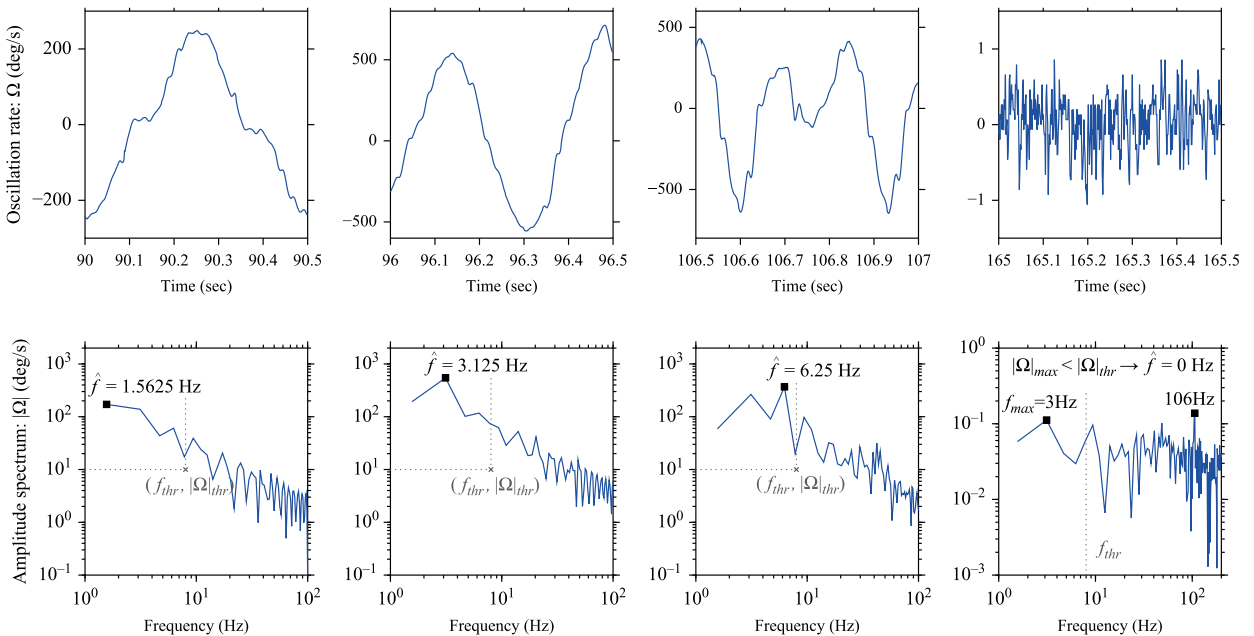


Fig. 5. Demonstration of the use of a FFT-based frequency estimation algorithm on four different data packets.

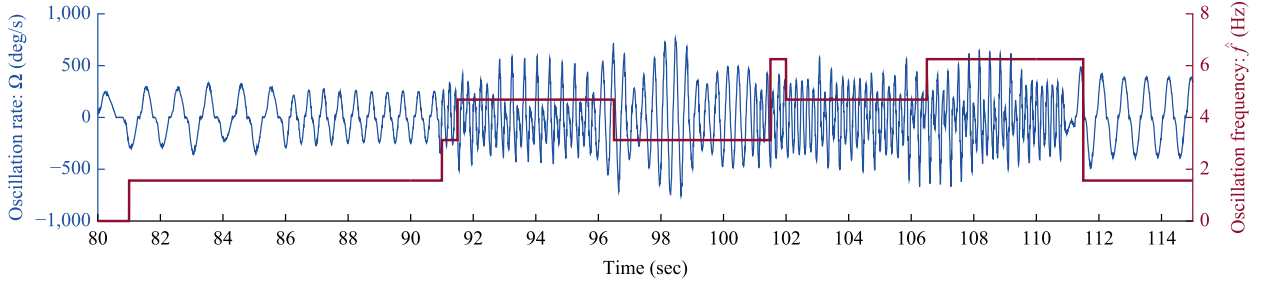


Fig. 6. Real-time FFT-based oscillation frequency estimation.

5.2. Measuring external acceleration magnitude

The magnitude of the external acceleration can be derived from the accelerometer measurements. If the magnitude of the accelerometer measurement is approximately equal to the gravitational acceleration (i.e., $\sqrt{A_x^2 + A_y^2 + A_z^2} \approx 1 \text{ g}$), then the system is in a non-accelerating mode. Therefore, it is common practice to measure the external acceleration using a switching model in which different threshold levels are assigned to the dynamic acceleration $\alpha = \left| \sqrt{A_x^2 + A_y^2 + A_z^2} - 1 \text{ g} \right|$. However, it is rather difficult to select and distinguish the appropriate threshold levels. Consequently, the defined threshold levels can easily result in a false external acceleration. Moreover, the scalar dynamic acceleration α provides brief and instantaneous results that do not provide an overall picture of the dynamics of the system.

The proposed approach takes advantage of the high sampling frequency of the employed accelerometer and formulates an accumulated measure to describe the magnitude of the external acceleration. The accumulated measure given by Eq. (23) utilizes a window of length L and integrates the instantaneous scalar dynamic acceleration. Therefore, the average acceleration provides a broad description of the dynamic behavior of the system. The window size can be varied based on the application design requirements, and, for shorter delays, smaller L values can be chosen. For simplicity, we chose $L = 400$ to synchronize the measures. The average acceleration can be calculated by:

$$\hat{\alpha} = \frac{1}{L} \sum_{k=1}^L \alpha_k, \quad \alpha_k = \left| \sqrt{A_{x,k}^2 + A_{y,k}^2 + A_{z,k}^2} - 1 \text{ g} \right|, \quad (23)$$

where $A_{x,k}$, $A_{y,k}$, and $A_{z,k}$ are the measurements of the three-axis accelerometer. Therefore, similar to the vibration measurement, the magnitude of the external acceleration $\hat{\alpha}$ is determined by collecting data packets of length L from the accelerometer and calculating the mean value of the scalar dynamic acceleration.

Fig. 7 depicts the determination of the external acceleration magnitude over a section of the 350s measurement. The blue curve shows the real external acceleration (α) applied to the test bench and measured by the plate encoder, while the red curve highlights the average dynamic acceleration ($\hat{\alpha}$) determined by Eq. (23). The average dynamic acceleration describes the dynamical behavior of the system accurately.

5.3. Fuzzy inference machine

Using the measures $\hat{f}$ and $\hat{\alpha}$, we can make deductions relevant to the dynamical behavior of the system. Hence, the measurement noise variance $R = \rho$ can be manipulated according to these parameters. The gyro-related noise is not sensitive to

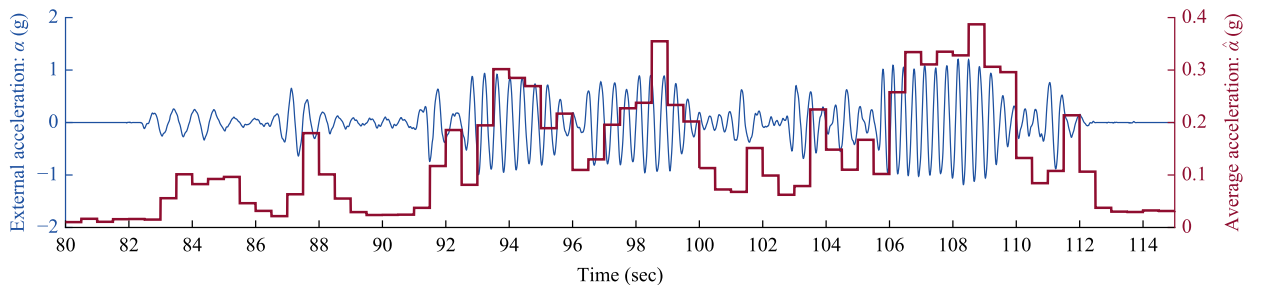


Fig. 7. Real-time determination of the external acceleration magnitude.

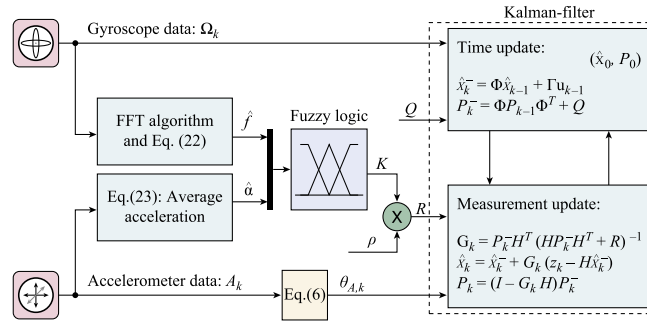


Fig. 8. Structure of the adaptive Kalman filter approach.

either the external acceleration or the vibrations (i.e., considering constant process noise covariance $Q = \text{diag}(q_{00}, q_{11})$). Consequently, an adaptive filtration is established to vary the noise variance ρ as a function of the measures $\hat{f}$ and $\hat{a}$.

Fuzzy reasoning enables deductions to be made using simple IF-THEN linguistic rules. Because fuzzy sets are applied, there is no need for complex mathematical relations. Instead, mapping between $\hat{f}$, $\hat{a}$ and the noise variance ρ can be performed using heuristic knowledge. The algorithm is composed of three main steps: the fuzzification of crisp inputs, fuzzy output calculations based on empirical IF-THEN rules, and the defuzzification of the fuzzy output [61].

The empirical IF-THEN rules are usually aggregated based on both observations related to the system and human common sense. In our case, the initial deductions consisted of two points:

- IF the system is in a stationary (non-accelerating) mode, THEN there exists a well-chosen ratio between the noise covariance values Q and R yielding a satisfactory state estimation performance, whereas
- IF external disturbances are present during the dynamical behavior, THEN (it is expected that the attitude realization $\theta_{A,k}$ is characterized by high uncertainty, and therefore) the measurement noise variance ρ should be increased so that the KF relies more heavily on the gyroscope data.

Fig. 8 depicts the structure of the proposed adaptive KF. A two-input one-output fuzzy inference machine with Mamdani architecture was implemented. The machine's inputs were the external disturbance magnitudes $\hat{f}$ and $\hat{a}$, while its output was a weighting factor denoted by K that weighted the noise variance (i.e., the adaptive measurement noise variance was formulated as $R = K\rho$). The ranges of the input variables were defined based on the fact that the maximum values of $\hat{f}$ and $\hat{a}$ were about 6.5 Hz and 0.35 g, respectively, while the output range was chosen intuitively via iterative tuning. The inputs were covered by three triangular membership functions where Z (zero), S (small), and B (big) describe the magnitudes of $\hat{f}$ and $\hat{a}$. In terms of the output, five singleton consequents ($K_1..K_5$) representing the scaling magnitude were defined. Fig. 9 shows the input and output membership functions and the properties of the fuzzy inference machine, while Fig. 10 depicts the fuzzy surface generated.

The aforementioned initial deductions were expanded into nine rules. These rules describe the scaling of the noise variance ρ according to the magnitudes of both the instantaneous vibration and the external acceleration as follows.

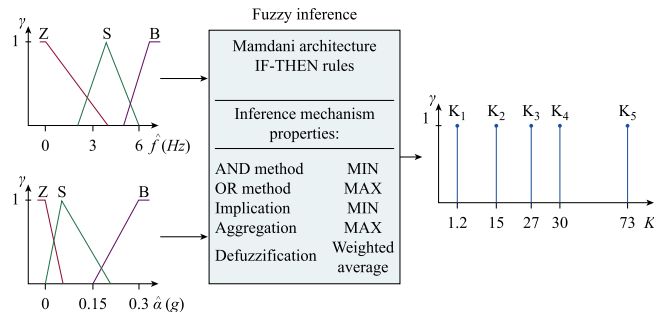


Fig. 9. Properties of the applied fuzzy inference machine.

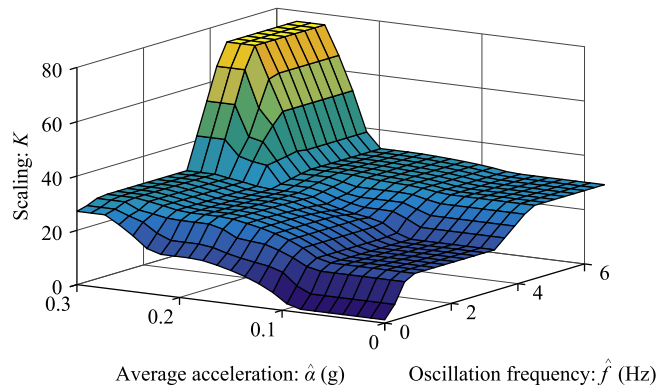


Fig. 10. Generated surface related to the fuzzy rule base.

- Rule1 : IF $\hat{f}$ is Z and $\hat{\alpha}$ is Z THEN K is K_1
 Rule2 : IF $\hat{f}$ is Z and $\hat{\alpha}$ is S THEN K is K_2
 Rule3 : IF $\hat{f}$ is Z and $\hat{\alpha}$ is B THEN K is K_3
 Rule4 : IF $\hat{f}$ is S and $\hat{\alpha}$ is Z THEN K is K_2
 Rule5 : IF $\hat{f}$ is S and $\hat{\alpha}$ is S THEN K is K_3
 Rule6 : IF $\hat{f}$ is S and $\hat{\alpha}$ is B THEN K is K_4
 Rule7 : IF $\hat{f}$ is B and $\hat{\alpha}$ is Z THEN K is K_3
 Rule8 : IF $\hat{f}$ is B and $\hat{\alpha}$ is S THEN K is K_4
 Rule9 : IF $\hat{f}$ is B and $\hat{\alpha}$ is B THEN K is K_5
- (24)

The rule base is summarized in Table 2. For example, IF the vibration is close to zero and the external acceleration is big, THEN the moderate scaling K_3 is applied, while IF both the vibration and the external acceleration are big, THEN the largest scaling K_5 is chosen.

Since the fuzzy architecture executes weighted average defuzzification, therefore the resulting crisp output (weighting factor K) and the adaptive measurement noise variance (R) are given as:

$$K = \frac{\sum_{i=1}^9 \kappa^i \cdot \min(\gamma^i(\hat{f}), \gamma^i(\hat{\alpha}))}{\sum_{i=1}^9 \min(\gamma^i(\hat{f}), \gamma^i(\hat{\alpha}))},$$

$$R = K\rho,$$

(25)

where $\gamma^i(\hat{f})$ and $\gamma^i(\hat{\alpha})$ are the i th-rule fired membership function values and κ^i denotes the singleton value of the consequent (K) of the i th rule (see Fig. 9).

5.4. Results

A flexible, adaptive filter structure was established based on the implementation of the aforementioned fuzzy inference machine. However, the noise covariance values Q and R had to be reselected by taking into account the fact that the measurement noise covariance R scales according to the dynamical behavior. Since both the optimization environment and the fitness function already had been established, it was feasible to re-execute the optimization for Q and R in the adaptive structure.

Table 2

Rule base of the fuzzy inference machine.

Scaling factor K		Ext. accel. $\hat{\alpha}$		
		Z	S	B
Vibration $\hat{f}$	Z	K_1	K_2	K_3
	S	K_2	K_3	K_4
	B	K_3	K_4	K_5

The bold letters indicate the fuzzy sets.

Table 3

Initial and optimized values and optimization bounds (for the adaptive case).

Symbol	Initial	Optimized	min	max
<i>First run: $F_{init} = 1.7596 \rightarrow F_{opt} = 1.6995$</i>				
ρ	0.05	0.14805	0.005	0.2
q_{00}	$7 \cdot 10^{-8}$	$2.71 \cdot 10^{-7}$	$1 \cdot 10^{-8}$	$3 \cdot 10^{-7}$
q_{11}	$5 \cdot 10^{-12}$	$3.09 \cdot 10^{-12}$	$2 \cdot 10^{-12}$	$5 \cdot 10^{-11}$
<i>Second run: $F_{init} = 1.7054 \rightarrow F_{opt} = F_{adapt} = 1.6990$</i>				
ρ	0.14	0.31969	0.05	0.43
q_{00}	$3 \cdot 10^{-7}$	$5.86 \cdot 10^{-7}$	$6 \cdot 10^{-8}$	$8 \cdot 10^{-7}$
q_{11}	$4 \cdot 10^{-12}$	$6.43 \cdot 10^{-12}$	$1 \cdot 10^{-12}$	$1 \cdot 10^{-11}$

The bold values indicate the optimized filter parameters.

Table 3 illustrates the optimization results. All of the initial noise variance values were set based on the results of Section 4.4, except for the variance ρ , which was initiated with a smaller value since it scales with the fuzzy output K according to the fuzzy surface depicted in Fig. 10. The optimization algorithm was executed twice in succession. The optimized variances are highlighted in the third column of Table 3. The ratio between the noise variances decreased (compared to the results in Table 1) due to the algorithm's adaptive characteristics. The algorithm's inherent flexibility allowed the process noise variance q_{00} to converge to a value ten times larger than in the previous case, resulting in faster estimation dynamics, whereas the process noise variance q_{11} converged to a notable bigger value as well. This outcome was expected, because the fuzzy scaling caused measurement noise variance ρ to increase each time a disturbance occurred, and, via this mechanism, the reliability of the accelerometer-based results was controlled real-time.

Both the adaptive KF approach and the optimized noise variances further improved the state estimation performance (i.e., the fitness function value settled at $F_{adapt} = 1.6990$). Compared to the optimized $F_{opt} = 1.9077$ fitness function value from Section 4, the adaptive KF approach improved the overall filtration performance by 10.9%. These results demonstrate that a superior filter convergence can be achieved by varying the noise variances according to the magnitudes of external disturbances.

6. Comparison

This section presents a comparison of the KF and adaptive KF approaches presented in Sections 4 and 5, respectively, against the commonly used orientation estimation methods proposed by Mahony et al. [62] and Madgwick et al. [63]. These filters (hereinafter Mahony and Madgwick filters) have gained extensive interest in the robotics and control community [64,65] and their performance is regularly taken into account as a benchmark in comparative analyses [66–68]. To consider the best performances in this analysis and conduct a fair comparison, the parameters of both filters were optimized in the same environment using the same fitness function, data set (sensor data and sampling time), and optimization procedure as discussed in Section 4. The implementation of the Mahony and Madgwick filters was based on the sample codes provided in [69].

6.1. Optimized Mahony filter

Complementary filters use frequency domain information (instead of statistical descriptions) to filter and combine signals provided by sensors that have complementary spectral characteristics. This also allows fast response and accuracy in orientation estimation.

Mahony et al. [62] formulated the filtering problem as a deterministic observation problem posed directly on the special orthogonal group $SO(3)$ driven by reconstructed attitude and angular velocity measurements. As a result, an explicit complementary filter was proposed that provides good orientation and gyro bias estimates based on accelerometer and gyroscope data. This nonlinear quaternion-based complementary filter first calculates the orientation error using the accelerometer data and the orientation determined in the previous step; then, a proportional and integral (PI) controller is employed to correct the gyroscope measurement. Through the integration of the quaternion propagation and normalization a new estimate of orientation is obtained.

For our tests, the adjustable K_p, K_I parameters of the PI controller were initially set to 0.5 and 0, respectively; then, the optimization of these parameters was executed using Eq. (23) and the PSO algorithm with $n_{gen} = 20$ and $n_{pop} = 20$. The results in Table 4 demonstrate that the optimization noticeably improved the filter performance; i.e., the initial fitness function value $F_{init} = 2.0042$ was reduced to $F_{opt} = 1.7849$ with the tuned K_p, K_I parameters.

6.2. Optimized Madgwick filter

The Madgwick filter also uses a quaternion representation of orientation. Its specificity lies in the application of accelerometer data in an analytically derived and optimized gradient descent algorithm to compute the direction of the

Table 4

Initial and optimized values and optimization bounds (for the Mahony filter).

Symbol	Initial	Optimized	min	max
<i>First run: $F_{init} = 2.0042 \rightarrow F_{opt} = 1.8276$</i>				
K_p	0.5	0.3502	0.35	0.7
K_I	0	$0.993 \cdot 10^{-4}$	0	0.005
<i>Second run: $F_{init} = 1.8274 \rightarrow F_{opt} = 1.7849$</i>				
K_p	0.35	0.2613	0.125	0.4
K_I	$1 \cdot 10^{-4}$	$2.3158 \cdot 10^{-4}$	0	$4 \cdot 10^{-4}$

The bold values indicate the optimized filter parameters.

Table 5

Initial and optimized values and optimization bounds (for the Madgwick filter).

Symbol	Initial	Optimized	min	max
<i>First run: $F_{init} = 2.8091 \rightarrow F_{opt} = 2.1228$</i>				
β	0.1	0.0371	0.02	0.15
<i>Second run: $F_{init} = 2.1333 \rightarrow F_{opt} = 2.1206$</i>				
β	0.035	0.0387	0.01	0.06

The bold values indicate the optimized filter parameters.

gyroscope measurement error as a quaternion derivative. The output of this algorithm yields a drift corrective step that maintains the gyro data-based quaternion propagation.

The initial value of the filters adjustable parameter β was set to 0.1; then, optimization was performed using Eq. (23) and the PSO algorithm with $n_{gen} = 10$ and $n_{pop} = 10$. Table 5 summarizes the outcome of the optimization, where the optimized $\beta = 0.0387$ is actually quite close to the value recommended by Madgwick et al. [63]. The executed optimization improved the fitness function value from the initial $F_{init} = 2.8091$ to $F_{opt} = 2.1206$.

6.3. Results

Based on the optimization results, the largest fitness function value ($F_{opt} = 2.1206$) corresponded to the performance of the Madgwick filter (the smaller the value, the better the performance). This drawback is related to the filter's constant gain property, meaning that it is unable to adapt to dynamic circumstances and modify its parameter based on the magnitudes of both the instantaneous vibration and the external acceleration. The KF discussed in Section 4 provided a slightly more robust estimation performance. The filter parameters were also constant values, but the combination of the optimized noise variances (ρ , q_{00} and q_{11}) resulted in a slightly better fitness function value ($F_{opt} = 1.9077$). The advantage of the KF can be related to its higher flexibility (three filter parameters) and its state-space model based property, which is characterized by noise statistics. The Mahony filter displayed the second-most competitive performance with a $F_{opt} = 1.7849$ fitness function value. This nonlinear complementary filter overcame the effect of dynamic motion and disturbances despite its constant gain property. The well-tuned proportional and integral controller (K_p and K_I parameters) allowed it to achieve satisfactory filter performance. However, the most robust filter performance in the high accelerating and vibrating test environment was attained by the adaptive KF discussed in Section 5 ($F_{opt} = 1.699$). The improved fit is achieved through the application of adaptive gains that are modified according to the perceived external disturbances.

In order to show the generality and robustness of the obtained adaptive KF, the performances of the analyzed filters were evaluated on four independent measurements (Measurement 1–4 lasted for 120, 170, 155, 150 s, respectively). The executed measurements are characterized by the presence of magnitudes of external accelerations, angular velocities and oscillation frequencies. The dynamic circumstances in which the filter performance was investigated are depicted on normalized histograms in Fig. 11. The first row shows the presence of different external acceleration magnitudes. It can be observed, that in approximately 65% of every measurement, external acceleration was applied in the range (0, 3.85] g, where the most intense circumstances appear in the fourth measurement (indicated by the purple curve in Fig. 11). The second row illustrates that angular velocities varied in the range (0, 735] deg/s in about 80% of each measurement. Finally, the third row shows the different inner body oscillation frequencies, that were present during the measurements. It can be seen that, in the range of [0, 9] Hz, the analyzed oscillation frequencies were present roughly in the same ratio.

Table 6 summarizes the filters' performances based on the mean squared error (MSE) and standard deviation (STD) of the attitude estimation error results, proving that the adaptive KF introduced in Section 5 outperforms the other filtering methods in each measurement. The results in Table 6 validate the performance of the proposed filtering approach. Nevertheless, it is worth mentioning that the generality and flexibility of this adaptive KF allows for further improvements. Some potential improvements are as follows.

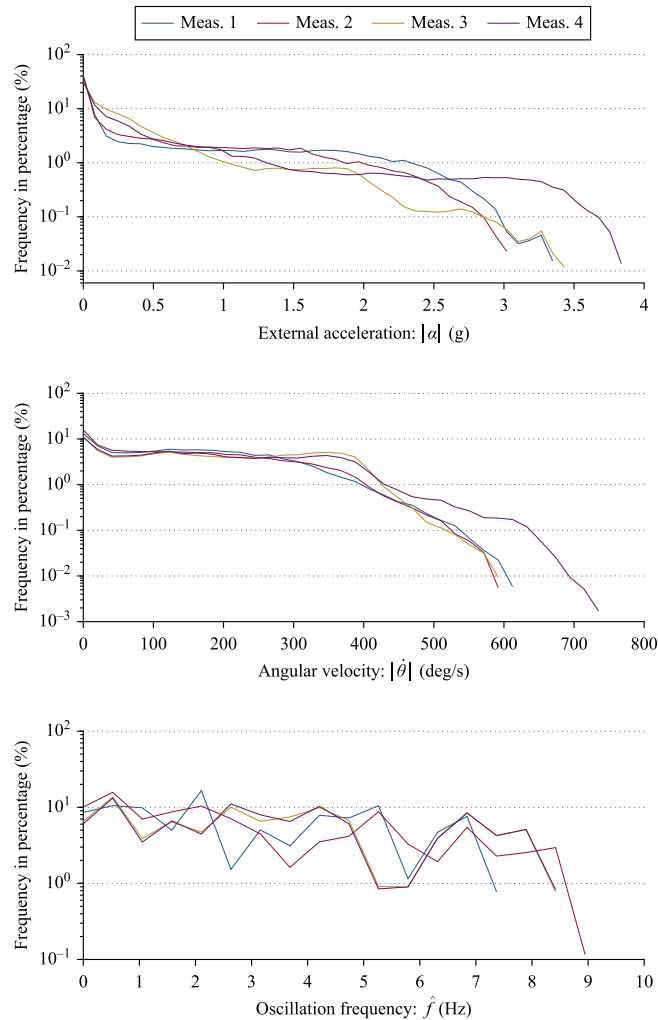


Fig. 11. Characterization of the executed measurements.

1. Employing a more sophisticated fuzzy inference machine in which the fuzzy input–output ranges are partitioned into additional fuzzy sets (resulting in an advanced rule base).
2. Optimizing the shapes and ranges of the membership functions and the weights of the applied rules.
3. Extending the fuzzy inference machine with an additional output (two-input two-output fuzzy machine) that also weights the noise variances of the gyro measurements.
4. Varying the window size in the calculation of external disturbance magnitudes in order to obtain more precise estimates of the vibration frequency and the average external acceleration.
5. Extending the filter structure with additional sensor information, e.g., with a tri-axis magnetometer.
6. Employing an acceleration model in the state space equations (e.g., similar to [26]), where the driving noise varies based on the disturbance magnitudes.

These issues are left open for investigation in future studies. We demonstrated that the methods for measuring external disturbance magnitudes provide relevant system behavior information. These methods can be applied to any motorized robotic system (e.g., one involving UAVs). Similarly, vibrations and external acceleration are the two primary sources of disturbance. We also demonstrated that fuzzy logic provided a simple, expert-oriented solution to establishing complex relations between the aforementioned disturbances and the filter parameters by formulating a set of heuristic IF-THEN rules. In our case, the KF was based on a simple two-dimensional state-space model of IMU data in which the measurement noise variance was manipulated based on the system behavior. Both the proposed fuzzy inference machine and the proposed disturbance measurement methods can be used to tune other filters in real time. This means that novel adaptive (and nonlinear) complementary filters (e.g., similar to [64]) can be formed and their performances can be investigated for different

Table 6
MSE and STD results of the investigated filters.

Filter	Measurement 1		Measurement 2		Measurement 3		Measurement 4	
	MSE	STD	MSE	STD	MSE	STD	MSE	STD
KF	1.5525	1.2452	1.7138	1.3089	2.2762	1.4860	2.6376	1.4754
Madgwick	2.1219	1.3316	4.1010	1.8919	2.4844	1.5762	2.6097	1.6017
Mahony	1.5772	1.1694	1.7591	1.2295	2.0404	1.4283	2.3340	1.5080
Adaptive KF	1.4310	1.1896	1.5485	1.2160	1.9109	1.3810	2.2614	1.4357

The bold values highlight the best performances.

mechatronic applications. Additionally, the proposed disturbance measurement methods can be employed in the elaboration of adaptive control (e.g., adaptive PID) solutions.

7. Conclusion

This paper proposes two novel solutions for low-cost MEMS-IMU based mobile-robot attitude estimation. First, we introduced a test bench that allows for simulation of various (accelerating and non-accelerating) system behaviors, and collecting measurements related to the true system states along with the IMU data. The collected measurements were used in a simulation environment, and the optimal KF parameters were determined by formulating a performance index and utilizing the PSO algorithm. This optimization solution was able to replace ad hoc KF tuning and improve the filtration performance by 78% over the initial filter convergence. The second part of the paper investigated the performance of an adaptive KF structure in which the filter parameters varied in real time based on the instantaneous system dynamics. These dynamics were characterized by the magnitudes of the external acceleration and vibrations, and two measurement approaches (FFT-based oscillation frequency determination and average acceleration calculation) were proposed for determining these magnitudes. The mapping between the instantaneous dynamics and the KF parameters was conducted by fuzzy-logic based heuristic IF-THEN rules. Compared to the optimized KF, the proposed adaptive approach improved the overall filter performance by 10.9%. The filter performances were benchmarked against two popular IMU-based algorithms with the aim to indicate that the proposed methods can provide robust performances and even outperform them. The adaptive KF approach could be advantageous for any mechatronic system in which variable operating conditions are present and the filter parameters are dynamic-dependent. Future work will involve both the extension of the proposed adaptive KF to three dimensions as well as the performance evaluation of the proposed techniques on a real two-wheeled inverted pendulum robot.

Acknowledgments

This work/publication was supported by the EFOP-3.6.1-16-2016-00003 project. The project is co-financed by the European Union. The authors would also like to thank Dr. Fritz Faulhaber GmbH & Co. KG for providing the high precision DC micromotors that were used in the mobile inverted pendulum system.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.ymssp.2018.03.053>.

References

- [1] I. Kecskés, E. Burkus, F. Bazsó, P. Odry, Model validation of a hexapod walker robot, *Robotica* 35 (2) (2017) 419–462.
- [2] Á. Odry, J. Fodor, P. Odry, Stabilization of a two-wheeled mobile pendulum system using LQG and fuzzy control techniques, *Int. J. Adv. Intell. Syst.* 9 (1.2) (2016) 223–232.
- [3] Á. Odry, E. Burkus, P. Odry, LQG Control of a two-wheeled mobile pendulum system, *The Fourth International Conference on Intelligent Systems and Applications (INTELLI 2015)*, 2015a, pp. 105–112.
- [4] F. Dai, X. Gao, S. Jiang, W. Guo, Y. Liu, A two-wheeled inverted pendulum robot with friction compensation, *Mechatronics* 30 (2015) 116–125.
- [5] H. Lee, S. Jung, Balancing and navigation control of a mobile inverted pendulum robot using sensor fusion of low cost sensors, *Mechatronics* 22 (1) (2012) 95–105.
- [6] C.-H. Huang, W.-J. Wang, C.-H. Chiu, Design and implementation of fuzzy control on a two-wheel inverted pendulum, *IEEE Trans. Industr. Electron.* 58 (7) (2011) 2988–3001.
- [7] Y.S. Suh, Attitude estimation by multiple-mode Kalman filters, *IEEE Trans. Ind. Electron.* 53 (4) (2006) 1386–1389.
- [8] C. Mazza, M. Donati, J. McCamley, P. Picerno, A. Cappozzo, An optimized Kalman filter for the estimate of trunk orientation from inertial sensors data during treadmill walking, *Gait Posture* 35 (1) (2012) 138–142.
- [9] S. Zhang, S. Yu, C. Liu, X. Yuan, S. Liu, A dual-linear Kalman filter for real-time orientation determination system using low-cost MEMS sensors, *Sensors* 16 (2) (2016) 264.
- [10] H.J. Luinge, P.H. Veltink, Measuring orientation of human body segments using miniature gyroscopes and accelerometers, *Med. Biol. Eng. Comput.* 43 (2) (2005) 273–282.

- [11] M. Zhong, J. Guo, Q. Cao, On designing PMI Kalman filter for INS/GPS integrated systems with unknown sensor errors, *IEEE Sens. J.* 15 (1) (2015) 535–544.
- [12] S. Zihajezadeh, D. Loh, T.J. Lee, R. Hoskinson, E.J. Park, A cascaded Kalman filter-based GPS/MEMS-IMU integration for sports applications, *Measurement* 73 (2015) 200–210.
- [13] M.F. Abdel-Hafez, K. Saadeddin, M.A. Jarrah, Constrained low-cost GPS/INS filter with encoder bias estimation for ground vehicles applications, *Mech. Syst. Signal Process.* 58 (2015) 285–297.
- [14] G. Reina, M. Paiano, J.-L. Blanco-Claraco, Vehicle parameter estimation using a model-based estimator, *Mech. Syst. Signal Process.* 87 (2017) 227–241.
- [15] H. Nourmohammadi, J. Keighobadi, Fuzzy adaptive integration scheme for low-cost SINS/GPS navigation system, *Mech. Syst. Signal Process.* 99 (2018) 434–449.
- [16] Y. Liu, X. Fan, C. Lv, J. Wu, L. Li, D. Ding, An innovative information fusion method with adaptive Kalman filter for integrated INS/GPS navigation of autonomous vehicles, *Mech. Syst. Signal Process.* 100 (2018) 605–616.
- [17] C.J. Allred, D. Churchill, G.D. Buckner, Real-time estimation of helicopter rotor blade kinematics through measurement of rotation induced acceleration, *Mech. Syst. Signal Process.* 91 (2017) 183–197.
- [18] T.D. Powell, Automated tuning of an extended Kalman filter using the downhill simplex algorithm, *J. Guidance, Control, Dynam.* 25 (5) (2002) 901–908.
- [19] O.V. Korniyenko, M.S. Sharawi, D.N. Aloï, Neural network based approach for tuning Kalman filter, in: *Electro Information Technology, 2005 IEEE International Conference on*, IEEE, 2005, pp. 1–5.
- [20] C. Kownacki, Optimization approach to adapt Kalman filters for the real-time application of accelerometer and gyroscope signals' filtering, *Digital Signal Process.* 21 (1) (2011) 131–140.
- [21] N. Salvatore, A. Caponio, F. Neri, S. Stasi, G.L. Cascella, Optimization of delayed-state Kalman-filter-based algorithm via differential evolution for sensorless control of induction motors, *IEEE Trans. Industr. Electron.* 57 (1) (2010) 385–394.
- [22] K. Shi, T. Chan, Y. Wong, S. Ho, Speed estimation of an induction motor drive using an optimized extended Kalman filter, *IEEE Trans. Industr. Electron.* 49 (1) (2002) 124–133.
- [23] M. Wang, Y. Yang, R.R. Hatch, Y. Zhang, Adaptive filter for a miniature MEMS based attitude and heading reference system, in: *Position Location and Navigation Symposium, 2004. PLANS 2004*, IEEE, 2004, pp. 193–200.
- [24] W. Li, J. Wang, Effective adaptive Kalman filter for MEMS-IMU/magnetometers integrated attitude and heading reference systems, *J. Navigat.* 66 (01) (2013) 99–113.
- [25] D. Jurman, M. Jankovec, R. Kamnik, M. Topič, Calibration and data fusion solution for the miniature attitude and heading reference system, *Sensors Actuators A: Phys.* 138 (2) (2007) 411–420.
- [26] J.K. Lee, E.J. Park, S.N. Robinovitch, Estimation of attitude and external acceleration using inertial sensor measurement during various dynamic conditions, *IEEE Trans. Instrument. Meas.* 61 (8) (2012) 2262–2273.
- [27] J. Gośliński, M. Nowicki, P. Skrzypczyński, Performance comparison of EKF-based algorithms for orientation estimation on Android platform, *IEEE Sens. J.* 15 (7) (2015) 3781–3792.
- [28] M. Nowicki, J. Wietrzykowski, P. Skrzypczyński, Simplicity or flexibility? Complementary Filter vs. EKF for orientation estimation on mobile devices, in: *Cybernetics (CYBCONF)*, 2015 IEEE 2nd International Conference on, IEEE, 2015, pp. 166–171.
- [29] Z. Wu, J. Shi, X. Zhang, W. Ma, B. Chen, et al, Kernel recursive maximum correntropy, *Signal Process.* 117 (2015) 11–16.
- [30] B. Chen, X. Liu, H. Zhao, J.C. Principe, Maximum correntropy Kalman filter, *Automatica* 76 (2017) 70–77.
- [31] L. Xi, B. Chen, H. Zhao, J. Qin, J. Cao, Maximum Correntropy Kalman filter with state constraints, *IEEE Access* 5 (2017) 25846–25853.
- [32] M.V. Kulikova, Square-root algorithms for maximum correntropy estimation of linear discrete-time systems in presence of non-Gaussian noise, *Systems Control Lett.* 108 (2017) 8–15.
- [33] G.Y. Kulikov, M. Kulikova, Estimation of maneuvering target in the presence of non-Gaussian noise: A coordinated turn case study, *Signal Process.* 145 (2018) 241–257.
- [34] S. Habbachi, M. Sayadi, F. Fnaiech, N. Rezzoug, P. Gorce, M. Benbouzid, Estimation of IMU orientation using linear Kalman filter based on correntropy criterion, in: *IEEE International Conference on Industrial Technology (ICIT)*, IEEE, 2018, pp. 1340–1344.
- [35] J.L. Crassidis, F.L. Markley, Y. Cheng, Survey of nonlinear attitude estimation methods, *J. Guidance, Control, Dynam.* 30 (1) (2007) 12–28.
- [36] F.L. Markley, J.L. Crassidis, *Fundamentals of Spacecraft Attitude Determination and Control*, vol. 33, Springer, 2014.
- [37] J.-X. Xu, Z.-Q. Guo, T.H. Lee, Design and implementation of a Takagi–Sugeno-type fuzzy logic controller on a two-wheeled mobile robot, *IEEE Trans. Ind. Electron.* 60 (12) (2013) 5717–5728.
- [38] F. Höflinger, J. Müller, R. Zhang, L.M. Reindl, W. Burgard, A wireless micro inertial measurement unit (IMU), *IEEE Trans. Instrum. Meas.* 62 (9) (2013) 2583–2595.
- [39] G. Welch, G. Bishop, An introduction to the Kalman filter, *Tech. Rep. TR 95-041*, Department of Computer Science, University of North Carolina, USA, 2001.
- [40] Á. Odry, I. Harmati, Z. Király, P. Odry, Design, realization and modeling of a two-wheeled mobile pendulum system, *14th International Conference on Instrumentation, Measurement, Circuits and Systems (IMCAS '15)*, 2015b, pp. 75–79.
- [41] Á. Odry, E. Burkus, I. Kecskés, J. Fodor, P. Odry, Fuzzy control of a two-wheeled mobile pendulum system, in: *Applied Computational Intelligence and Informatics (SACI)*, 2016 IEEE 11th International Symposium on, IEEE, 2016b, pp. 99–104.
- [42] Á. Odry, Kalman filter supplementary material, Available from: www.appl-dsp.com, URL <http://appl-dsp.com/lqg-and-fuzzy-control-of-a-mobile-wheeled-pendulum/>, 2018.
- [43] N. El-Sheimy, H. Hou, X. Niu, Analysis and modeling of inertial sensors using Allan variance, *IEEE Trans. Instrument. Meas.* 57 (1) (2008) 140–149.
- [44] A. Benini, R. Senatore, F. D'Angelo, D. Orsini, E. Quattraro, M. Verola, A. Pizzarulli, A closed-loop procedure for the modeling and tuning of Kalman Filter for FOG INS, in: *Inertial Sensors and Systems Symposium (ISS)*, 2015 DGON, IEEE, 2015, pp. 1–19.
- [45] H. Modares, A. Alfi, M.-M. Fateh, Parameter identification of chaotic dynamic systems through an improved particle swarm optimization, *Expert Syst. Appl.* 37 (5) (2010) 3714–3720.
- [46] S.-K. Oh, H.-J. Jang, W. Pedrycz, A comparative experimental study of type-1/type-2 fuzzy cascade controller based on genetic algorithms and particle swarm optimization, *Expert Syst. Appl.* 38 (9) (2011) 11217–11229.
- [47] Á. Odry, I. Kecskés, E. Burkus, Z. Király, P. Odry, Optimized Fuzzy Control of a Two-Wheeled Mobile Pendulum System, *Int. J. Control Syst. Rob.* 2 (2017) 73–79.
- [48] Á. Odry, I. Kecskés, E. Burkus, P. Odry, Protective fuzzy control of a two-wheeled mobile pendulum robot: design and optimization, *WSEAS Trans. Syst. Control* 12 (2017) 297–306.
- [49] A. Chatterjee, K. Pulasinghe, K. Watanabe, K. Izumi, A particle-swarm-optimized fuzzy-neural network for voice-controlled robot systems, *IEEE Trans. Industr. Electron.* 52 (6) (2005) 1478–1489.
- [50] G. Alici, R. Jagielski, Y.A. Şekericioğlu, B. Shirinzadeh, Prediction of geometric errors of robot manipulators with Particle Swarm Optimisation method, *Rob. Autonom. Syst.* 54 (12) (2006) 956–966.
- [51] N. Kwok, Q. Ha, T. Nguyen, J. Li, B. Samali, A novel hysteretic model for magnetorheological fluid dampers and parameter identification using particle swarm optimization, *Sensors Actuators A: Phys.* 132 (2) (2006) 441–451.
- [52] H. Modares, A. Alfi, M.-B.N. Sistani, Parameter estimation of bilinear systems based on an adaptive particle swarm optimization, *Eng. Appl. Artif. Intell.* 23 (7) (2010) 1105–1111.
- [53] G. Quaranta, G. Monti, G.C. Marano, Parameters identification of Van der Pol–Duffing oscillators via particle swarm optimization and differential evolution, *Mech. Syst. Signal Process.* 24 (7) (2010) 2076–2095.

- [54] A. Alireza, PSO with adaptive mutation and inertia weight and its application in parameter estimation of dynamic systems, *Acta Automat. Sin.* 37 (5) (2011) 541–549.
- [55] A. Alfi, M.-M. Fateh, Intelligent identification and control using improved fuzzy particle swarm optimization, *Expert Syst. Appl.* 38 (10) (2011) 12312–12317.
- [56] E.M. Hendrix, G. Boglárka, *Introduction to Nonlinear and Global Optimization*, Springer, New York, 2010.
- [57] I. Kecskés, P. Odry, Optimization of PI and Fuzzy-PI controllers on simulation model of Szabad (ka)-II walking robot, *Int. J. Adv. Rob. Syst.* 11 (11) (2014) 186.
- [58] Y. Ye, C.-B. Yin, Y. Gong, J.-j. Zhou, Position control of nonlinear hydraulic system using an improved PSO based PID controller, *Mech. Syst. Signal Process.* 83 (2017) 241–259.
- [59] J.F. Kennedy, J. Kennedy, R.C. Eberhart, Y. Shi, *Swarm Intelligence*, Morgan Kaufman, 2001.
- [60] G. Code, Particle swarm toolbox for MATLAB (psomatlab), Available from: [code.google.com](https://code.google.com/p/psomatlab/), URL <https://code.google.com/p/psomatlab/>, 2013.
- [61] L.-X. Wang, *A Course in Fuzzy Systems and Control*, Prentice-Hall PTR, Englewood Cliffs, NJ, 1997.
- [62] R. Mahony, T. Hamel, J.-M. Pflimlin, Nonlinear complementary filters on the special orthogonal group, *IEEE Trans. Automatic Control* 53 (5) (2008) 1203–1218.
- [63] S.O. Madgwick, A.J. Harrison, R. Vaidyanathan, Estimation of IMU and MARG orientation using a gradient descent algorithm, in: *Rehabilitation Robotics (ICORR)*, 2011 IEEE International Conference on, IEEE, 2011, pp. 1–7.
- [64] M. Euston, P. Coote, R. Mahony, J. Kim, T. Hamel, A complementary filter for attitude estimation of a fixed-wing UAV, in: *Intelligent Robots and Systems, 2008. IROS 2008. IEEE/RSJ International Conference on*, IEEE, 2008, pp. 340–345.
- [65] N.G. Tsagarakis, D.G. Caldwell, F. Negrello, W. Choi, L. Baccelliere, V. Loc, J. Noorden, L. Muratore, A. Margan, A. Cardellino, et al, WALK-MAN: a high-performance humanoid platform for realistic environments, *J. Field Robot.* 34 (7) (2017) 1225–1259.
- [66] A. Cavallo, A. Cirillo, P. Cirillo, G. De Maria, P. Falco, C. Natale, S. Pirozzi, Experimental comparison of sensor fusion algorithms for attitude estimation, *IFAC Proc. Vol.* 47 (3) (2014) 7585–7591.
- [67] R.G. Valenti, I. Dryanovski, J. Xiao, Keeping a good attitude: A quaternion-based orientation filter for IMUs and MARGs, *Sensors* 15 (8) (2015) 19302–19330.
- [68] Q. Mourcou, A. Fleury, C. Franco, F. Klopčič, N. Vuilleme, Performance evaluation of smartphone inertial sensors measurement for range of motion, *Sensors* 15 (9) (2015) 23168–23187.
- [69] S. Madgwick, An efficient orientation filter for inertial and inertial/magnetic sensor arrays, Report x-io and University of Bristol (UK), vol. 25, 2010.